FROM THE

EDITOR-IN-CHIEF

Long before Covid-19 entered our lexicon, cancer was the Big C. And long after that virus of unknown origin slips into history—or becomes a tame omnipresence like the common cold—the mysteries of cancer will still loom menacingly in our midst. In fact, all through the pandemic, the enigmatic crab was silently exacting its toll. In 2020, nearly 800,000 people died of cancer in India—almost seven times those felled by Covid. It is those numbers that give us cause for Code Red-level alarm and vigil. For, though cancer is by no means a purely modern affliction, its rising graph in our times depressingly resembles that of an epidemic.

Through lived or anecdotal experience, most Indians already know the bad news. But here are some broad, and grim, numbers to help map the contours of this malignant force. Cancer killed 382,000 people in India in 1990. By 2016, that had more than doubled. Absolute numbers related to the annual increase in disease burden show a symmetry with that, having gone up from 548,000 in 1990 to 1.16 million in 2016. If you count patients who had lived with the disease for over five years, that year counted 2.26 million. Indeed, the numbers seem to start doubling at each layer of projection—almost like cancer cells multiplying. By 2025, India will have nearly 30 million cases, or about 2 per cent of our entire population.



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Till recently, there were two reasons that those diagnosed with cancer treated the news as being akin to a death sentence. One, while the pathologies related to cancer are well-mapped, its birth and causation still lie somewhat beyond the grasp of even the most advanced science. Even if many genetic and environmental triggers stand identified, that grim moment in the drama when the body gives up on its normal auto-regulatory function is still enacted on a largely unlit part of our inner stage. That is why the disease evokes the fear of the unknown—as much as that of mortality. Secondly, as immunotherapist Rahul Purwar, trained in this new form of precision oncology from Harvard, points out, "Till a decade ago, there were only three options for patients—chemotherapy, radiation or surgery. If these didn't work, you had no other choice—cancer was your death sentence."

But now, thanks to Purwar and others of his ilk, there is cause for cheer. When Purwar returned to India in 2013, he found that immunotherapy was entirely missing from India's arsenal of defences. Since then, he and a host of others have worked to change one fundamental fact: the disease being synonymous with a failure of the body's self-healing mechanisms. They have brought about a revolutionary shift in treatment by retooling that same healing faculty. Immunotherapy zeroes in on the body's own fighting force—and harnesses it for the good fight.

Kiran Mazumdar-Shaw, chairperson of Biocon, who along with famed author and researcher Siddhartha Mukherjee co-founded Immuneel, an Indian immunotherapy company, in 2019, is working to cut down the cost of such cancer treatment in India by as much as 300 per cent. Shaw points out, "From surgeries, chemotherapy and radiotherapy to antibody immunotherapies that target certain kinds of cancer antigens, we have come a long way." She talks of how the survival rates of some cancers have shot up—breast cancer, for example, had a survival rate of 10 per cent a decade ago; it is almost 90 per cent today if diagnosed in time. "The prognosis for some cancers is so good that you could say you might die with that cancer but you won't die of it."

That is why immunotherapy and a handful of allied or parallel advances truly come like light at the end of the tunnel. As senior associate editor Sonali Acharjee ranges through the most exciting kinds of cellular intervention for our cover story this week, it becomes more and more clear that these are guided by a level of understanding not attained before. Even in the middle of pitched battle, only an overall sense of the theatre can be a guarantor of eventual victory. This is beginning to happen. What used to be practically a death-knell upon its announcement now looks increasingly like any other disease: potentially deadly, but also curable.

Purwar's patented method—CD-19 CAR-T, India's first indigenous chimeric antigen receptor T-cell therapy—is awaiting deployment on the frontlines too. Gearing up for its Phase 2/ 3 trials, he hopes to offer the treatment in Indian hospitals by early next year. This procedure involves taking the human T cell—a white blood cell involved in immune responses—and training it to fight cancerous cells. This kind of intervention scripts a marvellous twist in nature's story, because cancer happens precisely when abnormally multiplying cells manage to do so by evading detection by the body's immune mechanism.

India now has several such notable ventures for immunotherapy, including Curadev, Bugworks Research, Immuneel Therapeutics, Zumutor and Aurigene Discovery Technologies. The focus is as much on reducing costs as on fine-tuning therapy. And investments are pouring in. Shaw and Mukherjee's Immuneel Therapeutics announced its Series A fundraising round of $15 million last month. Meanwhile, Bugworks raised $18 million for the pre-clinical development of molecules that will target tumours where the compound adenosine, a chemical found in human cells, plays a role in neurotranmission. Zumutor is filing an investigational new drug development with the US Food and Drug Administration for its NK Cell Therapy molecule (similar to CAR-T therapy, but with the focus on NK or 'natural killer' cells and not T cells) this year and recently raised $6.2 million in Series A4 funding.

We are now looking at a future of customised cancer cure, governed by a much more precise micro-level grasp of the bodily processes the disease entails. The full panoply of newer generation gene- and cell-based therapies include not only targeted interventions using small molecules and monoclonal antibodies, molecular oncology is now able to identify specific features in tumour biomolecules—DNA, RNA and protein expression. The last is crucial, as it relates to the very mechanism by which cancer evades detection by the body's radars. As new forms of diagnostics shine a torch into the dark, therapy is getting as granular with things like treatment vaccines. As of now, much of the advances are limited to a few forms of carcinoma, but humanity has surely crossed a frontier. The crab does not let go easily, but science is tightening its grip too.

(Aroon Purie)